

Middle Layers Know Where to Click: Coordinate Serialization Bottlenecks in GUI Agents

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Abstract

Autoregressive GUI agents ground instructions by generating screen coordinates, conflating two computations that the final-layer reading leaves undistinguished: locating the target widget in screen space, and serializing that location into coordinate tokens. We show these computations peak at different transformer depths: spatial grounding is best decoded from intermediate layers and often degrades near the output, while coordinate-token predictability keeps improving in later layers. We call this mismatch the **coordinate serialization bottleneck**. To exploit it, we introduce **ZWERGE-UI**, a frozen-backbone retrofit that reads intermediate-layer posteriors through lightweight cross-attention probes ($\approx 42\text{M}$ parameters, 0.5% of an 8B backbone), fuses them with a $<200\text{K}$ -parameter head, and routes each sample between the fused point and the backbone’s own coordinate via a training-free confidence gate. Experiments across 4 GUI agents, 2 backbone families, and 5 benchmarks show that the gate yields small but consistent and statistically significant gains (paired $p < 0.001$) under the identical strict success-rate protocol, at the cost of a single forward pass.

1 Introduction

Autoregressive GUI agents ground instructions by generating screen coordinates, conflating two computations: *locating* the target widget in screen space and *serializing* that location into a coordinate vocabulary. Prior work has exploited attention maps and prefill-stage signals to aid localization (Xu et al. 2025; Wu et al. 2025a; Lin et al. 2026), while analyses of vision transformers show that final-layer representations may lose spatial fidelity (Darcet et al. 2024; Lan et al. 2024; Chen et al. 2025). However, these lines do not ask *where within the prefill computation* spatial grounding is most accessible. At the same time, GUI-agent training studies reveal a tension between chain-of-thought reasoning and precise localization (Xu et al. 2026; Yang et al. 2026; Zhou et al. 2025; Zhang, Gao, and Xu 2025), motivating perception–reasoning decoupling (Zhang et al. 2026). We operationalize this by freezing the backbone and reading spatial beliefs directly from intermediate activations.

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We study this question with lightweight frozen-backbone readouts and reveal a consistent non-monotonic trajectory: spatial grounding improves through early and middle layers, peaks at intermediate depths, and often degrades toward the final layer. A complementary *serialization lens*—teacher-forced coordinate-token likelihood without additional parameters—shows that coordinate tokens become most predictable in later layers. This mismatch, which we call the **coordinate serialization bottleneck**, is not a generic saliency effect: an instruction-switch counterfactual shifts the mid-layer posterior from one target to another, and the bottleneck is most consequential in high-resolution layouts with small targets, where final-layer accuracy is lowest.

Motivated by this, we introduce **ZWERGE-UI**, a frozen-backbone retrofit for existing GUI agents. **ZWERGE-UI** attaches lightweight cross-attention probes to intermediate layers to recover layerwise spatial posteriors, combines them with a small cross-layer fusion head, and then applies a training-free confidence gate that decides, per sample, when to trust the intermediate-layer point over the backbone’s own autoregressive coordinate. The probes serve a dual role: they provide the analytical lens that reveals the bottleneck, and they form the grounding component used at inference. Importantly, **ZWERGE-UI** does not modify the agent backbone or its reasoning pathway.

We evaluate on four GUI agents spanning two backbone families, Qwen2.5-VL and Qwen3-VL, across five GUI grounding benchmarks, ScreenSpot-Pro, ScreenSpot-v2, OSWorld-G, MMBench-GUI, and UI-Vision, six configurations in total since we evaluate OSWorld-G on both its refusal-removed and original splits. The strongest spatial readout consistently appears before the final layer. A training-free confidence gate converts this into consistent grounding gains across all four backbones and six benchmarks (§5.2). The gains are small in magnitude but statistically significant: under a paired test on the samples where the gate and the native decoder disagree, the ensemble is net-positive in the large majority of cells and the pooled test over all samples is decisive ($p < 0.001$); no cell is significantly worse than the baseline (supplementary). We read the result as evidence that intermediate-layer signals can complement native decoding, not as a new state-of-the-art grounding system.

In summary, our contributions are:

- We identify the **coordinate serialization bottleneck**: intermediate layers encode more accurate instruction-conditioned spatial posteriors than final layers, while later layers better predict coordinate tokens.
- We introduce two complementary diagnostic lenses—a frozen-backbone *spatial lens* and a parameter-free *serialization lens*—and show a robust depth mismatch across four GUI agents.
- We present ZWERGE-UI, a lightweight frozen-backbone retrofit that recovers intermediate-layer spatial signals through cross-attention probes and a training-free, GT-free confidence gate that routes each sample between the fusion point and the native autoregressive coordinate. Across two backbone families and six benchmarks, all four retrofitted agents improve over their native baselines under the identical strict success-rate protocol, at near-zero added cost.

2 Per-Layer Spatial Probes

We diagnose *where* GUI grounding signals reside across transformer depth by attaching one lightweight probe per layer to a frozen backbone. The same probe set later forms the readout head of the retrofit (§4). Probes introduce no new spatial knowledge: the backbone is frozen, and the gradient from layer ℓ reaches only ϕ_ℓ .

Prefill-forced inference. Coordinate generation serializes a location into up to eight number tokens, but the spatial intent is already encoded before decoding begins (§3). We prefill a fixed template ending in an anchor token, `click(<|ground|><|pointer_start|>...)`, and run a single forward pass. At the anchor position the model has processed the full screenshot and instruction but no coordinate value, so there is zero label leakage.

Cross-attention probe. Let f_θ be a frozen L -layer backbone. For each probed layer $\ell \in \mathcal{L}_{\text{probe}}$, drawn from the deeper half of the network where spatial signal concentrates, we read the anchor query $\mathbf{h}_{t^*}^{(\ell)} \in \mathbb{R}^d$ at the `<|ground|>` position and visual keys $\mathbf{H}_v^{(\ell)} \in \mathbb{R}^{N_v \times d}$ over the N_v patch tokens. Each probe ϕ_ℓ computes a patch-level posterior via multi-head cross-attention with a learnable head gate α :

$$\mathbf{Q}_\ell = W_q^{(\ell)} \text{Norm}(\mathbf{h}_{t^*}^{(\ell)}), \quad \mathbf{K}_\ell = W_k^{(\ell)} \text{Norm}(\mathbf{H}_v^{(\ell)}),$$

$$p_\ell = \sigma \left(\sum_h \sigma(\alpha)_h \frac{\mathbf{Q}_{\ell,h}^\top \mathbf{K}_{\ell,h}}{\sqrt{d_h}} \right) \in \Delta^{N_v-1}, \quad (1)$$

where $\text{Norm}(\cdot)$ is the RMS pre-scale followed by LayerNorm, and $W_q^{(\ell)}, W_k^{(\ell)}$ are full-rank projections so each probe learns a geometry suited to its layer. Each probe is $\approx 4.2\text{M}$ parameters ($d=4096, n_h=8, d_h=64$); the full set totals $\approx 42\text{M}$, or 0.52% of a 7B backbone.

RMSNorm pre-scale. Hidden-state ℓ_2 norms grow with depth, geometrically misaligning layers. We pre-scale $\mathbf{h} \leftarrow \mathbf{h}/(\|\mathbf{h}\|_2/\sqrt{d})$ before each LayerNorm, bounding the input norm to $\approx \sqrt{d}$ regardless of depth and preventing bfloat16 overflow.

Supervision. Each probe is supervised with an anisotropic Gaussian centered at the bbox centroid:

$$y_i \propto \exp \left(-\frac{(x_i - c_x)^2}{2\sigma_x^2} - \frac{(y_i - c_y)^2}{2\sigma_y^2} \right), \quad \sigma_x = \eta w_b, \sigma_y = \eta h_b \quad (2)$$

which reduces over-penalization from patch-grid quantization on small targets (§3). Stage 1 minimizes $\mathcal{L}_{S1} = \frac{1}{|\mathcal{L}_{\text{probe}}|} \sum_\ell D_{\text{KL}}(\mathbf{y} \| p_\ell)$ with the backbone receiving zero gradient. Training-data construction is detailed in the supplementary material.

3 The Coordinate Serialization Bottleneck

We investigate *where* GUI grounding signals are encoded across transformer layers, progressing from the layerwise spatial profile (Finding 1), through the spatial–serialization dissociation (Finding 2) and its instruction-conditioning (Finding 3), to the resolution-dependent consequence (Finding 4).

Experimental setup. We study four GUI agents spanning two backbone families, Qwen2.5-VL (Qwen Team 2025a) and Qwen3-VL (Qwen Team 2025b), on ScreenSpot-Pro (Li et al. 2025) and ScreenSpot-v2 (Wu et al. 2025b). Spatial grounding quality is measured via a lightweight per-layer grounding head, Stage 1 of §2, which is under 1% of backbone parameters and is trained with the backbone completely frozen, so it introduces no new spatial knowledge. Coordinate serialization quality is measured via the backbone’s native language model head under teacher forcing, with no additional parameters.

3.1 Spatial Grounding Peaks at Intermediate Layers

Autoregressive GUI agents implicitly assume final-layer representations carry the richest spatial information. The blue curves in Figure 1 challenge this: spatial quality peaks at intermediate depths, then declines toward the output, a non-monotonic trajectory consistent across both backbone families.

Finding 1. *Intermediate transformer layers are consistently better spatial decoders than the final layer.*

Robustness. The inverted-U is robust to distribution and sample size: across all five benchmarks, L^* varies by ≤ 3 layers per model (std ≤ 1.1), and peak exceeds final in 20/20 cells. Bootstrap resampling (1000 resamples) places $L^\dagger$ within ≤ 1 -layer 95% CI for every model and L^* within ≤ 3 layers for three of four; $L^* < L^\dagger$ holds in all four. Full tables are in the supplementary material.

3.2 A Dissociation Between Spatial Localization and Coordinate Serialization

Finding 1 raises the question: if late layers are not spatial refiners, what do they do? We hypothesize they specialize in *coordinate serialization*—translating a latent spatial intent into the vocabulary-level token sequence. We test this with a *serialization lens*: coordinate-token log-likelihood measured via teacher forcing at each layer.

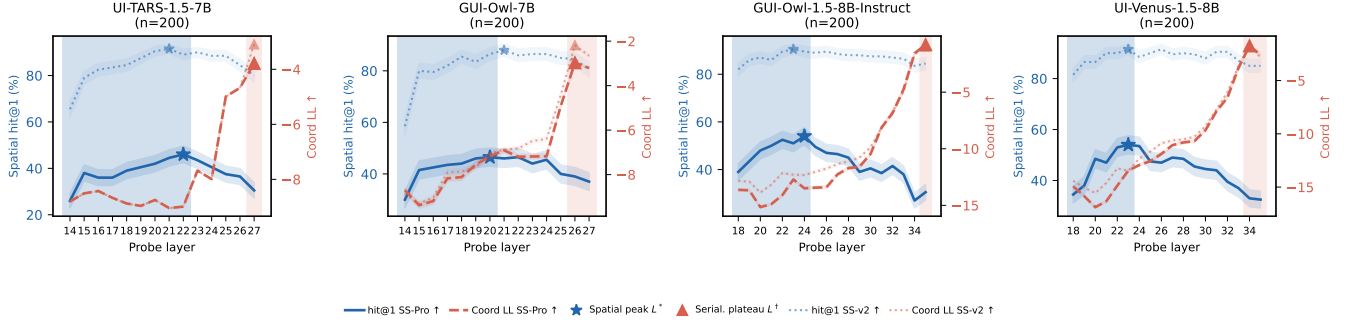


Figure 1: **Spatial localization and coordinate serialization peak at different depths.** Blue curves (left axis): spatial hit@1 $\uparrow$. Orange curves (right axis): coordinate token log-likelihood $\uparrow$. Both signals rise together, then diverge at the spatial peak L^* (blue star): spatial quality declines while serialization continues rising to $L^\dagger$ (orange triangle). The consistent lag $L^\dagger > L^*$ across all models constitutes direct evidence for the coordinate serialization bottleneck.

The orange curves in Figure 1 reveal a dissociation: both signals rise together through early layers, then diverge—spatial quality peaks and falls while coordinate log-likelihood continues climbing to a later plateau. The lag $L^\dagger > L^*$ is consistent across all four models and both backbone families, constituting direct evidence for a **coordinate serialization bottleneck**.

Finding 2. *The layer that best decodes the target region consistently precedes the layer that best predicts the coordinate token sequence. Late layers are coordinate serializers, not spatial refiners.*

Logit-lens controls. Applying the native LM head to intermediate states is a logit-lens-style probe (nostalgebraist 2020); late layers could align better with the output space for reasons unrelated to coordinates (Geva et al. 2021; Meng et al. 2022). Two controls address this; full tables are in the supplementary material.

Non-coordinate control. We repeat the teacher-forced measurement on *non-coordinate* protocol tokens (parentheses, JSON keys) from the same generations. These reach near-zero NLL at late layers for three of four models, confirming generic alignment. Crucially, coordinate *value* tokens retain a substantial residual NLL at the plateau (2.2–3.1 nats for Qwen2.5-VL; 1.0–1.7 nats for Qwen3-VL), even where format tokens are fully serialized. This residual is the coordinate-specific bottleneck component. The plateau $L^\dagger$ coincides for coordinate and protocol tokens in 3/4 models, so $L^\dagger$'s *location* is a general late-layer phenomenon while the coordinate-specific signal is the residual gap.

Tuned-lens control. Following (Belrose et al. 2023), we fit a per-layer affine map to the final-layer representation and apply the *frozen* output head. Calibration flattens the raw trend (early-layer NLL: 8.6 $\rightarrow$ 3.8 nats), showing part of the raw improvement is an alignment artifact. The plateau layer, however, is unchanged (L26 both before and after), so the *locus* is robust. Together, the residual NLL gap and

the calibration-invariant plateau indicate genuine coordinate serialization.

3.3 The Mid-Layer Posterior Is Instruction-Conditioned, Not Visual Saliency

An alternative is that mid-layer spatial signals merely reflect *visual saliency*. To test this, we design a counterfactual: for pairs sharing the same screen but targeting disjoint elements, we swap the instruction while keeping the image fixed.

Figure 2 shows the posterior shifts dramatically: switching the instruction collapses the spatial mass on the original target, concentrated at exactly the intermediate layers where grounding peaks. This rules out the saliency account.

Finding 3. *Swapping the instruction while keeping the image fixed causes a large shift in the mid-layer spatial posterior. The spatial signal is instruction-conditioned grounding, not visual saliency.*

3.4 The Bottleneck Is More Consequential on High-Resolution, Fine-Grained Layouts

Finally, we ask whether the bottleneck is more consequential where GUI grounding is hardest—high-resolution screen-shots with small, densely packed targets.

We quantify severity via the *collapse gap*: peak-layer hit@1 minus final-layer hit@1. Table 1 shows the gap is consistently larger on ScreenSpot-Pro than on ScreenSpot-v2 across every model.

Controlled within-benchmark target-size regression. SS-Pro vs SS-v2 conflates target size with resolution. To isolate size, we hold the benchmark fixed at ScreenSpot-Pro ($n=1581$) and bin by target bbox area (supplementary material). The peak-final *gap* does *not* increase monotonically as targets shrink: it rises for the two Qwen2.5-7B models but peaks at intermediate sizes for the Qwen3-8B models, floored by peak accuracy on sub-patch targets. Absolute final-layer hit@1, however, is $\sim 1.7\times$ worse on the smallest

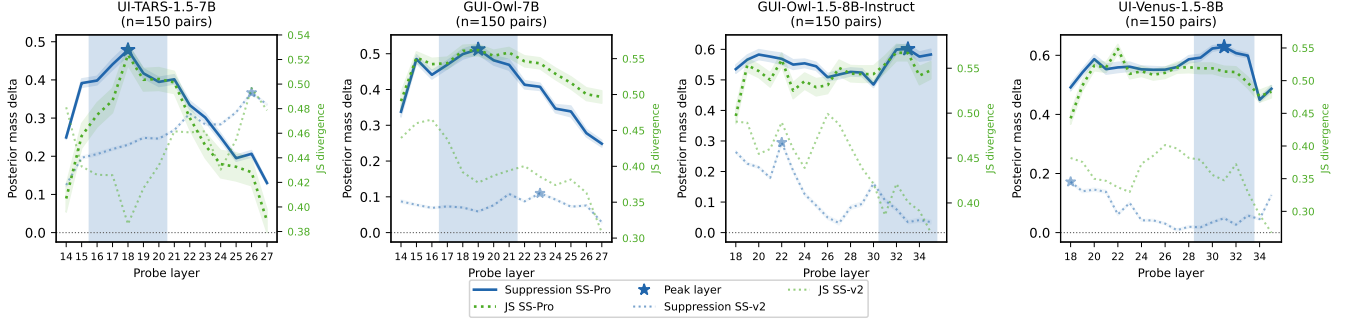


Figure 2: **Instruction-switch counterfactual: the mid-layer posterior is instruction-conditioned.** Blue curves: old-target suppression $\uparrow$. Orange curves: posterior JS divergence $\uparrow$. Both effects peak at the same intermediate layers as spatial grounding quality (cf. Fig. 1) and are substantially nonzero, ruling out visual saliency.

Model	SS-Pro			SS-v2		
	Pk _{probe}	Fin _{probe}	Gap $\downarrow$	Pk _{probe}	Fin _{probe}	Gap $\downarrow$
UI-TARS-1.5-7B	46.0	30.5	15.5	91.5	80.0	11.5
GUI-Owl-7B	46.5	37.0	9.5	88.0	80.5	7.5
GUI-Owl-1.5-8B	54.0	30.5	23.5	90.5	84.5	6.0
UI-Venus-1.5-8B	54.0	32.5	21.5	91.5	85.0	6.5

Table 1: **Collapse gap is larger on high-resolution layouts.** Pk_{probe} and Fin_{probe} are the peak-layer and final-layer *spatial-probe* hit@1 (pp) from §3.1, not the backbone’s native autoregressive accuracy reported as “Native” in Table 2; the two tables use different metrics (hit@1 here vs. overlap@1 there) and different readouts (probe vs. native decoder). SS-Pro gaps consistently exceed SS-v2 gaps.

quintile ($\sim 34\%$ vs $\sim 58\%$), so the bottleneck is most damaging in absolute terms on small targets even though the gap is not monotone in size.

Finding 4. *The collapse gap is larger on high-resolution ScreenSpot-Pro than on mid-resolution ScreenSpot-v2. Within ScreenSpot-Pro, final-layer accuracy is lowest on the smallest targets even though the collapse gap is not monotone in target size; the bottleneck is most consequential, though not largest in gap, for fine-grained layouts.*

Taken together, the four findings motivate reading the spatial posterior from intermediate layers and using cross-layer fusion to select the optimal depth per sample, as described below.

4 ZWERGE-UI

The per-layer probes of §2 are independent, and the best layer varies by sample (§3, Finding 1). We add two lightweight components on top of the frozen probes, shown in Figure 3: a cross-layer fusion head that selects the reliable depth per sample, and a training-free confidence gate

that decides when to trust the fused point over the backbone’s own decoder. Both reuse the single prefill of §2, so the retrofit adds negligible compute over the native baseline.

4.1 Cross-Layer Fusion

We draw active layers $\mathcal{L}_{\text{active}} \subset \mathcal{L}_{\text{probe}}$ from the spatial plateau identified in §3 and fuse their posteriors. A cross-layer LoRA refines each anchor query $\mathbf{h}_{t*}^{(\ell)}$ into $\tilde{\mathbf{q}}_{\ell}$; a cosine-meta scorer then assigns per-layer weights:

$$\omega_{\ell} = \text{softmax}(\alpha_{\ell} + \tau^{-1} \cos(\tilde{\mathbf{q}}_{\ell}, \mathbf{q}_{\text{meta}})), \quad p_{\text{final}} = \sum_{\ell \in \mathcal{L}_{\text{active}}} \omega_{\ell} p_{\ell}, \quad (3)$$

where α_{ℓ} is a sample-independent trainable prior on layer ℓ ’s average quality, $\mathbf{q}_{\text{meta}}$ is a global trainable meta-query serving as a soft prototype of the ideal grounding representation, and τ is a learnable temperature. Layers whose refined query aligns with $\mathbf{q}_{\text{meta}}$ receive higher weight for the current input. The LoRA contributes $\approx 200\text{K}$ parameters, two orders of magnitude below a single probe.

Stage 2 loads the Stage-1 probes; the backbone and inactive probes stay frozen, while the fusion head and the active probes remain trainable, with loss

$$\mathcal{L}_{\text{S2}} = D_{\text{KL}}(\mathbf{y} \| p_{\text{final}}) + \frac{\lambda}{|\mathcal{L}_{\text{active}}|} \sum_{\ell} D_{\text{KL}}(\mathbf{y} \| p_{\ell}), \quad \lambda = 0.2, \quad (4)$$

so the per-layer auxiliary term gives every active probe a direct gradient signal and no single layer is neglected.

4.2 Training-Free Gated Decoding

The fusion posterior p_{final} is patch-level, not a coordinate. In one prefill we read both the fused point $\mathbf{f}$, produced by a training-free posterior-to-point decoder detailed in the supplementary material, and the backbone’s native autoregressive coordinate $\mathbf{n}$, and route between them with a GT-free gate on the posterior sharpness: emit $\mathbf{f}$ when the margin $m = p_{\text{final}}^{(1)} - p_{\text{final}}^{(2)}$, the difference between the top-two single-patch probabilities, exceeds τ , otherwise emit $\mathbf{n}$. A sharp mode means the probe is confident in one location; a

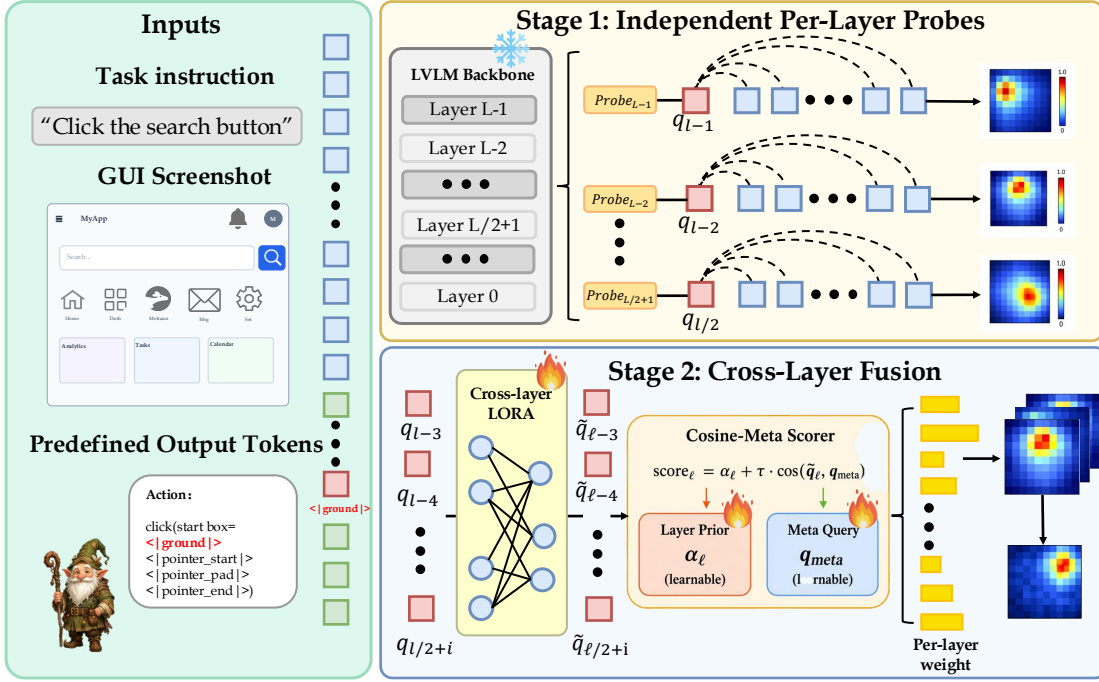


Figure 3: **Overview of ZWERGE-UI.** A frozen backbone exposes layerwise hidden states from a single prefill; independent probes decode intermediate-layer spatial posteriors, a cross-layer fusion head selects the reliable depth per sample, and a training-free confidence gate routes each sample between the fused point and the native autoregressive coordinate.

diffuse mode means the native serializer should be trusted. The default is $\tau=0.25$, while the main results use 0.20. This instantiates the paper’s thesis: the intermediate layer identifies *where* the target is, and a confidence gate decides when to trust that signal over the final-layer serializer.

Fairness and cost. The comparison against the native baseline is protocol-symmetric: both points are evaluated under the same strict point-in-box metric on the same cached run, and the gate uses no labels. The ensemble adds no autoregressive generation over the baseline. The native generate is the baseline’s own computation, and the fused point reads intermediate activations that the backbone already produces in that same forward; in principle the two share one prefill, so the full retrofit is realizable at the cost of a single forward pass.

5 Experiments

5.1 Experimental Setup

Benchmarks. We evaluate on six GUI grounding configurations: ScreenSpot-Pro (Li et al. 2025), ScreenSpot-v2 (Wu et al. 2025b), the original and refusal-removed OSWorld-G-refine splits of OSWorld-G (Xie et al. 2025), MMBench-GUI (Wang et al. 2025), and UI-Vision (Nayak et al. 2025), spanning desktop, web, and mobile.

Metrics. We report the strict single-point success rate (overlap@1); protocol and patch-mapping details are in the supplementary material.

Baselines. We retrofit four GUI agents: GUI-Owl-7B (Ye et al. 2025) and UI-TARS-1.5-7B (Qin et al. 2025), both built on the 28-layer Qwen2.5-VL, and GUI-Owl-1.5-8B (Xu et al. 2026) and UI-Venus-1.5-8B (Team et al. 2026), both built on the 36-layer Qwen3-VL. Each backbone is frozen throughout; only the probe and fusion heads are updated. The Native column is each backbone’s own autoregressive accuracy on the same task set, so the comparison isolates the retrofit.

Implementation. Stage 1 trains 14 (Qwen2.5-VL) or 18 (Qwen3-VL) CrossAttn probes per model on 200k grounding samples for one epoch; Stage 2 trains the fusion head with a subset of active probes, 10 layers for the Qwen2.5-VL backbones and 12 for the Qwen3-VL backbones, for one epoch, with the backbone and the remaining probes frozen. The deployed decoder is the margin-gated ensemble (§4.2) with $\tau=0.20$.

5.2 Main Results

Table 2 compares the confidence-gated ensemble against the native baseline under the identical strict success-rate protocol, with a single global threshold and no per-backbone tuning. Twenty of 24 cells improve and 21 of 24 are non-negative, for a macro-average gain of +0.46 pp; the three negative cells are all within 0.6 pp of zero. The gate routes to the fusion point on 5.7–32.7% of samples and otherwise defers to the native decoder; it fires most on OSWorld-G, where the native baseline has the most headroom, and least

Table 2: **Confidence-gated ensemble, strict overlap@1 (%)**. **Native**: backbone baseline. **+ZwerGe**: margin-gated ensemble (§4.2), same cached run. $\Delta = +\text{ZwerGe} - \text{Native}$. **%Fus.**: fraction routed to fusion. OSW-G = OSWorld-G (original); OSW-G-Ref = refusal-removed. Global $\tau=0.20$; 20/24 improve, 21/24 non-negative, macro-average +0.46 pp.

Backbone	Benchmark	Native ov@1	+ZwerGe ov@1	Δ	%Fus.
GUI-Owl-7B (Ye et al. 2025) Qwen2.5-VL	SS-Pro	55.22	56.10	+0.89	7.5
	SS-v2	92.84	93.08	+0.24	5.8
	OSW-G-Ref	79.61	80.39	+0.78	25.7
	OSW-G	72.55	72.94	+0.39	25.3
	MMBench-GUI	80.77	81.27	+0.50	7.5
	UI-Vision	53.87	54.42	+0.55	12.7
UI-TARS-1.5-7B (Qin et al. 2025) Qwen2.5-VL	SS-Pro	48.32	49.59	+1.27	6.8
	SS-v2	91.35	91.66	+0.31	5.7
	OSW-G-Ref	77.06	76.47	−0.59	24.5
	OSW-G	64.51	65.10	+0.59	20.8
	MMBench-GUI	77.46	77.88	+0.42	7.6
	UI-Vision	27.12	28.45	+1.33	7.6
GUI-Owl-1.5-8B (Xu et al. 2026) Qwen3-VL	SS-Pro	74.89	75.21	+0.32	10.2
	SS-v2	94.10	94.02	−0.08	9.9
	OSW-G-Ref	80.98	82.35	+1.37	32.7
	OSW-G	78.43	78.63	+0.20	30.4
	MMBench-GUI	83.58	84.17	+0.58	9.9
	UI-Vision	43.94	44.97	+1.04	14.2
UI-Venus-1.5-8B (Team et al. 2026) Qwen3-VL	SS-Pro	70.52	70.65	+0.13	7.7
	SS-v2	96.85	96.85	0.00	8.6
	OSW-G-Ref	86.08	86.27	+0.20	28.8
	OSW-G	81.76	82.35	+0.59	25.5
	MMBench-GUI	89.04	88.95	−0.08	8.8
	UI-Vision	52.18	52.33	+0.16	11.8

on ScreenSpot-v2, where native accuracy is near ceiling.

Interpreting the gains. We read these small, mostly positive gains as evidence that intermediate-layer signals can complement native decoding, not as a new state-of-the-art result. The magnitude is small by construction: the gate defers to the native decoder on the majority of samples and overrides only when its posterior is sharply peaked. On the samples where the two disagree, the ensemble is net-positive in the large majority of cells, and a pooled paired test over all samples is decisive in its favor ($p < 10^{-17}$); the gain is individually significant ($p < 0.05$) on the six highest-headroom cells, and no cell is significantly worse than the baseline (per-cell table in the supplementary).

5.3 Diagnosing the Gains: Target Geometry and Retrofit Headroom

We explain the two patterns in Table 2, the largest headroom on ScreenSpot-Pro and the smallest gains on Qwen3-VL, with two diagnostics computed from the same cached runs.

Why ScreenSpot-Pro is the hard case. Roughly three quarters of ScreenSpot-Pro targets are smaller than a single visual patch (see supplementary), so a patch-level posterior can at best identify the correct patch, not the sub-patch pixel. This is precisely why the retrofit is framed as a *gate* rather than a replacement decoder: the intermediate posterior identifies *which patch* contains the target, while sub-patch pre-

cision is left to the native serializer whenever the probe is uncertain.

Retrofit headroom predicts the gate’s behavior. Define the *retrofit headroom* $h = \max_{\ell} \text{ov}@1(p_{\ell}) - \text{ov}@1(\text{native})$. When $h > 0$, bypassing the serializer recovers real headroom; when $h \leq 0$, the native decoder has already closed that gap. A supplementary figure plots h against the ungated fusion-only delta. Qwen2.5-VL backbones, whose native decoders leave more headroom, occupy the positive-headroom, positive-gain quadrant; Qwen3-VL backbones, whose native decoders approach the intermediate-layer ceiling, sit near or below zero. This is the regime the confidence gate targets: it defers to the native decoder whenever the fusion posterior of Eq. 3 is diffuse, which is why most cells in Table 2 are non-negative.

6 Conclusion

We presented ZWERGE-UI, a frozen-backbone retrofit that exposes and exploits the coordinate serialization bottleneck in GUI agents: instruction-conditioned spatial posteriors peak at intermediate transformer layers, while later layers increasingly specialize in coordinate serialization rather than localization refinement. A training-free, GT-free confidence gate combines the fused intermediate-layer point with the native decoder at the cost of a single forward pass. Across two backbone families and six benchmarks, all four

retrofitted agents improve over their native baselines, and the improvement is statistically significant under a paired test ($p < 0.001$). The gains are modest by design: the retrofit’s purpose is to validate the bottleneck hypothesis rather than chase leaderboard scores, and the heads train on far less data than the backbone, leaving clear room for improvement as GUI agents scale.

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